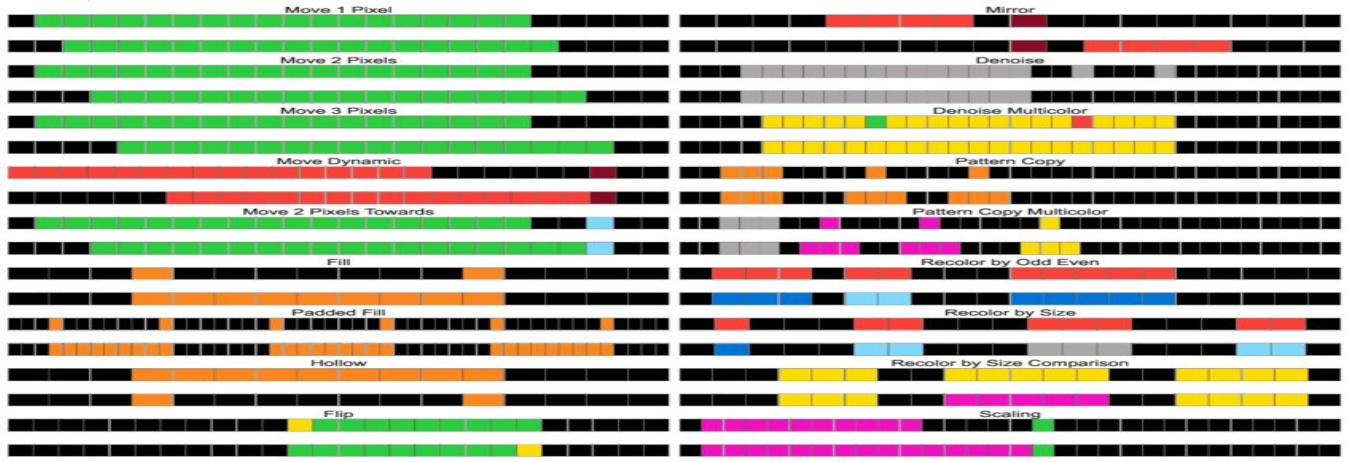
Solving 1D-ARC: A Meta-Neural Cellular Automata Approach

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Problem Formulation & Core Challenge

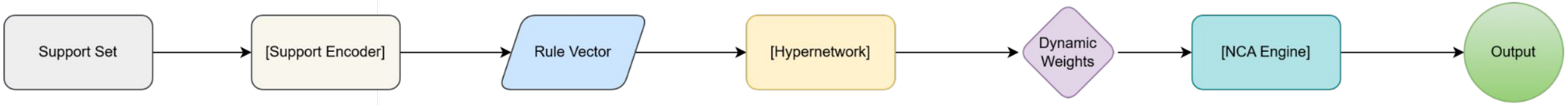


Objective: Design a fully offline, zero-shot system capable of few-shot algorithmic reasoning on the 1D-ARC dataset, strictly without external Large Language Models (LLMs).

The Twin Challenges of ARC:

- **Visual Overfitting:** Standard neural networks act as pattern matchers. They overfit to specific visual features (e.g., memorizing absolute color palettes) rather than deducing the underlying structural rules.
- **Out-of-Distribution (OOD) Extrapolation:** Generalizing to entirely unseen abstract concepts (Cross-Concept tasks) requires dynamic rule generation. Static weights fail when confronted with novel geometric logic.

Proposed Architecture: Meta-NCA



Modeling ARC tasks as iterative algorithmic transformations using a Meta-Learner that dynamically generates its own weights per task.

Pipeline Breakdown:

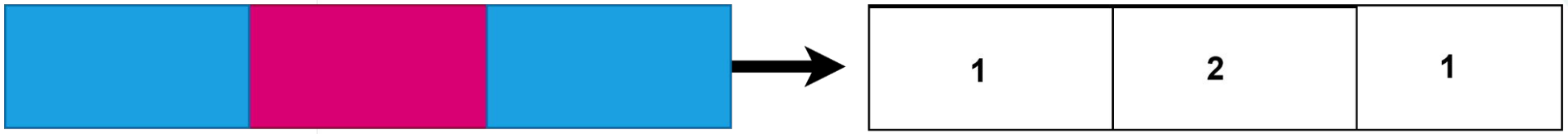
1. **Support Encoder:** Processes the few-shot input/output pairs to deduce the underlying algorithmic rule, compressing this logic into a dense, task-specific latent vector.
2. **Hypernetwork Bottleneck:** Acts as a dynamic rule-generator. It translates the latent vector into custom convolution weights and biases (FiLM/Direct Generation).
3. **Neural Cellular Automata (NCA) Engine:** Applies these dynamically generated, custom weights to the query input across multiple iterative steps to logically "compute" the final output.

Dynamic Abstraction

A "Role-Based Abstraction Layer" applied before data reaches the neural embedding to eliminate visual bias.

Mechanism & Impact:

- **How it Works:** Instead of feeding raw Color IDs (e.g., Red, Blue) to the model, colors are mapped dynamically based on their *appearance sequence*. The first color encountered becomes ID 1, the second becomes ID 2, etc.
- **Handling Novelty:** Unseen query colors or output-specific colors are safely assigned the next available IDs, preserving spatial logic.
- **The Result:** The model is forced to learn structural relationships (e.g., "Color 1 takes the properties of Color 2") rather than memorizing raw pixel values, rendering it completely immune to cross-task palette shifts.



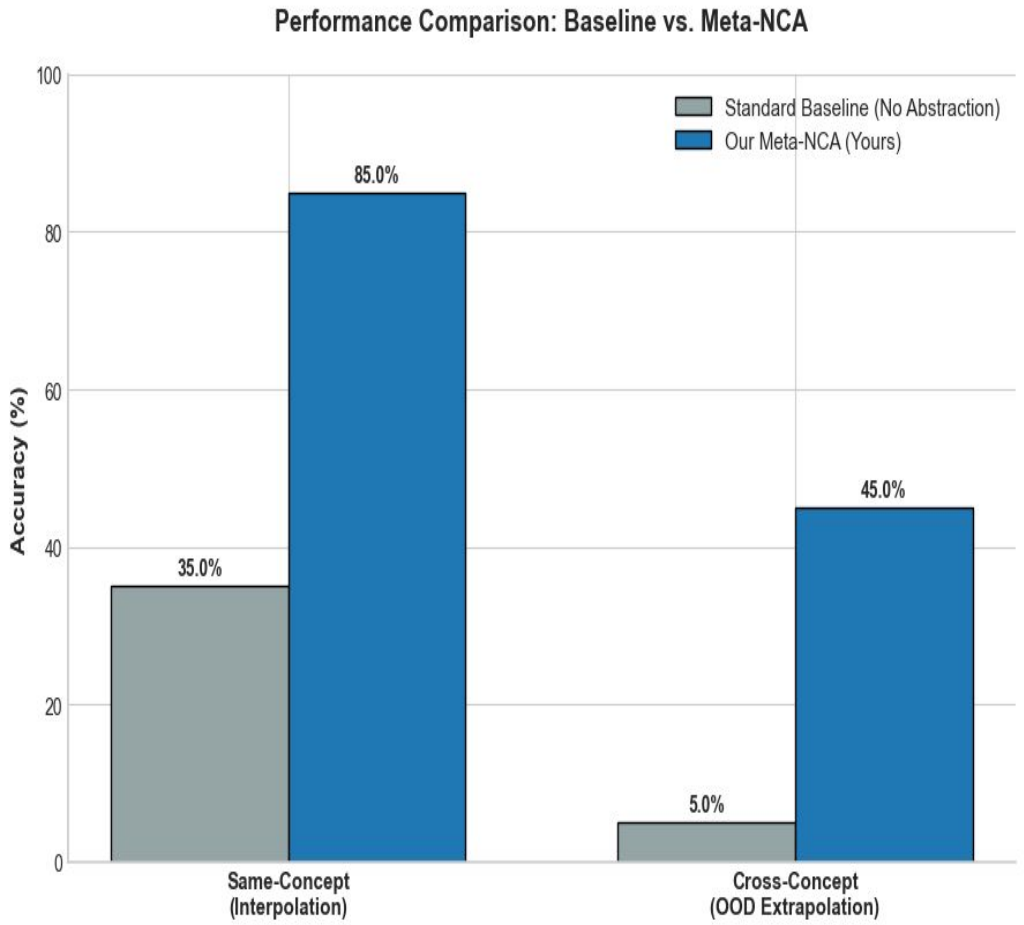
Experimental Setup & Results

Rigorous Evaluation Protocol:

- **Same-Concept:** 90/10 Train/Test split for standard interpolation.
- **Cross-Concept (Holdouts):** Strictly isolated three rule categories (`1d_scale_dp`, `1d_hollow`, `1d_recolor_cmp`) from training to rigorously test OOD extrapolation.

Current Baseline Metrics:

- **Same-Concept Accuracy (~85%):** Demonstrates strong interpolation and reliable rule-execution capability within the learned latent space.
- **Cross-Concept Accuracy (~45%):** Demonstrates the "Concept Distance" limitation. Pure neural models struggle to extrapolate to alien symbolic concepts without discrete program synthesis.



Next Steps for final submission

Next Steps for Pipeline Optimization:

- **Equivariant Data Augmentations:** Implementing strictly logic-preserving data augmentations (e.g., bidirectional spatial flipping) to increase spatial robustness without breaking ARC's absolute spatial anchors.
- **Hypernetwork Tuning:** Optimizing the latent bottleneck dimensionality to prevent catastrophic overfitting to the few-shot support set.
- **Addressing OOD Limitations:** Exploring hybrid methods or discrete search strategies to bridge the accuracy gap on extreme cross-concept tasks where continuous neural weights hit an extrapolation ceiling.